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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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D3.4 Risk based on the unknown unsafe scenarios

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Authors: Kevin Tom Kurian, Mohsen Alirezaei, Erjen Lefeber

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Project Members:

Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Technische Universiteit Eindhoven (TUE)

Kungliga Tekniska Hoegskolan (KTH)

Uppsala Universitet (UU)

Stichting Koninklijk Nederlands Lucht - En Ruimtevaartcentrum (NLR)

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List of Abbreviations

SBT Scenario-Based Testing

ADS Autonomous Driving System

SOTIF Safety of the Intended Functionality

ODD Operational Design Domain

ADS Autonomous Driving System(s)

RV Reference Vehicle

CM Criticality Measure

VOI Variables of Interest

SUT System Under Test

IDM Intelligent Driver Model

FOV Field of View

CG Center of gravity

1. Introduction

Building upon the mathematical framework for modeling scenarios in Deliverable D3.3, Deliverable D3.4 focuses on developing an appropriate Criticality Measure (CM) to assess scenarios. Quantifying the 'unsafeness' of a scenario using the CM forms an integral component of the cost function in the optimization process of WP5. The optimization in WP5 aims to minimize the risk from unknown-unsafe scenarios within the Operational Design Domain (ODD) of the Autonomous Driving System(s) (ADS), crucial for assessing the safety of the ADS in accordance with ISO 21448: Safety of the Intended Functionality (SOTIF) [2].

Scenario-Based Testing (SBT) is the most widely used method for safety verification and validation of Automated Driving System-equipped Vehicles (ADSs) [3]. To comply with ISO 21448, SBT can be formulated as an optimization problem to identify unknown hazardous scenarios within the ADS's ODD. We consider scenarios where the behavior of the ADS leads to high collision risk as hazardous/unsafe scenarios. Collision risk is defined as a combination of the probability and severity of harm in a scenario [2], and we consider collisions of the ADS to be harmful. The terms *risky*, unsafe, and *critical* are used interchangeably.

The work in [4] and [5] describes the optimization workflow for identifying unknown unsafe scenarios developed by Siemens in compliance with ISO 21448 for ground vehicles. In these works, each scenario is abstracted and described by a set of scenario parameters η. An objective function is formulated for the set of scenario parameters η, quantifying the scenario's probability of occurrence (Eq. 1.2) and criticality or 'unsafeness' (Eq. 1.3). This objective function from [5] is given as,

$$f(\eta) = f_1(\eta) f_2(\eta) \tag{1.1}$$

where,

$$f_1(\eta) = G(P_s(\eta)), \tag{1.2}$$
$$f_2(\eta) = 2 - \epsilon(\eta) + \mathrm{TTC}_{\min(\eta)}, \tag{1.3}$$

with,

$P_s(\eta)$ – Probability of occurence of scenario with parameter η.

$G(.)$ – Cost function dependent on the probability of scenario.

$\epsilon(\eta)$ – Unexpectancy factor for ego-vehicle in a scenario with parameter η.

$\mathrm{TTC}_{\min}(\eta)$ – Factor accounting for collision risk based on minimum TTC in scenario.

This objective function f is the cost function optimized to find the parameters associated with the

unknown-unsafe scenarios. The optimization problem is formulated in [5] as,

$$\begin{aligned} \min_{\eta} \quad & f(\eta) \\ \text{s.t.} \quad & \eta \in \chi \\ & \zeta_{\text{coll}} = 0, \end{aligned} \tag{1.4}$$

where χ is the set of valid scenario parameters and $\zeta_{\text{coll}} = 0$ is a boolean variable which is 1 in case of non-ego vehicle collisions.

However, the TTC-based CM used in the objective function, given in Eq. 1.3, is insufficient for adequately representing the risk for all scenarios. Certain requirements are to be satisfied by the CM for use in the aforementioned optimization framework. These requirements are given in Section 2.1. Several measures for scene/scenario criticality exist, and [6] briefly discusses the commonly used ones in detail in the context of these requirements. Overall, the work in [6], as part of D3.4, aims to develop a measure of collision risk/criticality for a scenario satisfying these requirements.

The key problem being addressed in [6] as part of Deliverable D3.4 is: What would be a suitable CM for scenarios, indicating collision risk for the ADS, for use as a cost function to select scenarios using optimization? The main contribution of this report is a measure of collision risk for the ADS in a scenario, including the probability and severity aspects. The measure is useful as an improved cost function for optimization-based scenario selection for SBT.

The structure of this report is as follows:

1. Chapter 2: Describes the requirements for the CM and discusses existing CMs.
2. Chapter 3: Describes the formulation and development of the proposed CM.
3. Chapter 4: Discusses the application of the CM to example scenarios.
4. Chapter 5: Concludes with suggestions for future work.

2. Evaluating existing criticality measures

The existing CMs in the literature are briefly evaluated against the requirements of the CM for use in the optimization framework. These requirements are listed below.

2.1 Requirements of the CM

The requirements of the CM taken from Kurian et al. [6], with an addition of R4 to search for high-risk scenarios, are given below:

R1: The aggregated CM $\mathfrak{C}$ should map the space of all scenarios in the optimization space $\mathfrak{B}$, to a non-negative real-number proportional to the scenario criticality, i.e., $\mathfrak{C} : \mathfrak{B} \rightarrow K$ where $K \subseteq [0, \infty)$.

R2: The CM should measure the criticality of near-misses to a collision of the ADS, i.e., the proximity of a scenario to a collision. This is necessary for the optimization to search for collision scenarios.

R3: The CM should serve the purpose of a cost function, i.e., the aggregated CM should

(a) Use information from the complete concrete scenario, i.e., the measure should be computed a posteriori.

(b) Be universally applicable for all scenarios within the optimization space. If the computation of $\mathfrak{C}$ depends on a reference trajectory/distribution, this reference should be commonly applicable for all scenarios within the optimization space.

R4: The CM should measure the collision risk of the ADS, i.e., include the probability and severity of a collision scenario. This is necessary to search for high-risk scenarios by excluding scenarios involving improbable, low severity collisions.

2.2 Existing criticality measures

Numerous measures for evaluating criticality exist in the literature, and an extensive review can be found in Westhofen et al. [7] and Singh et al. [8]. These CMs can be categorized based on the measured quantity, such as time, distance, acceleration, risk fields, and probability-based. In the context of requirements listed in Section 2.1, commonly used measures in literature are evaluated

in [6], with the conclusion that none of them satisfy all the requirements R1 to R4. In this report, a measure of proximity to collision is presented, which is based on the fraction of colliding futures and satisfies R1 to R4. The measure is based on the work in [6] and incorporates the severity and probability of collisions.

3. Defining the Criticality Measure (CM)

In this section, we define criticality/risk for a scenario in a general sense using the behavioral approach. We begin by briefly reiterating the definition of a scenario and related terminology.

3.1 Scenario definition: A behavioral approach

We use the Behavioral model-based definition of scenarios and scenario terminology from Kurian et al. [9], which is briefly summarized for completeness. Each concrete scenario ζ is considered a single trajectory of certain Variables of Interest (VOI) over a specific time interval. Given the domain for the VOI as $\mathbb{W}$ and a time vector $\mathbb{T} \subset \mathbb{R}^+ \cup \{0\}$, the space of all possible mappings/trajectories is called the universum $\mathbb{U} = \mathbb{T} \rightarrow \mathbb{W}$. Within this universum, the subset of all trajectories satisfying certain mathematical relations is called the behavior $\mathfrak{B}$. The CM maps each concrete scenario ζ to a real-valued output indicating the criticality of the concrete scenario. An example SBT optimization problem to find critical scenarios is formulated as,

$$\Psi = \arg_{\zeta \in \mathfrak{B}} \quad \mathfrak{C}(\zeta) > w. \tag{3.1}$$

where a CM $\mathfrak{C} : \mathfrak{B} \rightarrow [0, \infty)$ is defined to evaluate each scenario ζ such that the optimization problem is well-posed. Here, w is a threshold for criticality of scenarios that are considered interesting.

3.2 Criticality Measure (CM): A behavioral approach

To develop a measure evaluating the criticality of the ADS's behavior, we use a Reference Vehicle (RV) [6], which stays identical for all considered scenarios, making it a scenario-independent reference. Two possible choices for RV include the uncontrolled 'host vehicle' in the ADS or a commonly used vehicle without autonomous capability. We chose the host vehicle of the ADS as our RV, but the analysis remains similar for other choices.

In Kurian et al. [6], the hypothesis is that "the fraction of colliding trajectories in the set of feasible alternative futures for the RV tracing the ADS's trajectory indicates 'proximity to collision' of the ADS". However, we are interested in the 'proximity' to high-risk collisions only, i.e. collisions which are probable and severe, since the goal of the CM is to indicate high-risk scenarios. Considering the set of feasible futures for the RV as a sample space, and a random variable indicating the severity of each future, the collision risk is equivalent to the expected value of severity for the set. Hence, we posit that the safe driving behavior of an ADS minimizes the collision risk for an RV tracing the ADS's trajectory at any given time. Thus, the collision risk

for the RV indicates the criticality of ADS's behavior. Now, we formalize this using the behavioral modeling approach, similar to [6].

Consider that the VOI for this scenario includes the positions and velocities of both the ADS and the other actors. The System Under Test (SUT) is the ADS, specifically the ith ADS version combined with the host vehicle. We define $\mathfrak{B}_0$ in (3.2) as the set of all possible VOI trajectories that comply with the physical laws and constraints governing the RV. The set $\mathfrak{S}_i$ consists of VOI trajectories that adhere to the laws governing the ith ADS version. Similarly, $\mathfrak{A}_j$ represents the set of VOI trajectories that satisfy the behavioral laws of other actors and the initial condition specifications of the ego vehicle. The behavior of other actors and the ego vehicle's initial conditions are parameterized, where the subscript j denotes the jth set of parameter values. Thus, we define,

$$\mathfrak{B}_{i,0} = \mathfrak{B}_0 \cap \mathfrak{S}_i,\ \mathfrak{B}_{i,j} = \mathfrak{B}_{i,0} \cap \mathfrak{A}_j \text{ and } \mathfrak{B}_{0,j} = \mathfrak{B}_0 \cap \mathfrak{A}_j. \tag{3.2}$$

$\bigcup_{j\in J} \mathfrak{B}_{i,j} \subset \mathfrak{B}_{i,0}$ defines the optimization space in which we want to rate scenarios for safety-critical behavior of the SUT. J is the set of parameter values defining the actor behaviors that the SBT optimization aims to test the SUT against. Given the jth parameter value, $\mathfrak{B}_{i,j}$ and $\mathfrak{B}_{0,j}$ in (3.2) represent the combination of a specific behavior of actors and initial conditions with the SUT and the RV, respectively.

Next, for a given scenario $\zeta_{i,j} \in \mathfrak{B}_{i,j}$ and a behavior $\mathfrak{B}_{0,j}$, we define a restricted behavior $\mathfrak{B}_{0,j}(\zeta_{i,j}, [t_1, t_2])$ as,

$$\mathfrak{B}_{0,j}(\zeta_{i,j}, [t_1, t_2]) = \{\bar{\zeta} \in \mathfrak{B}_{0,j} s.t. \forall t \in [t_1, t_2] : \bar{\zeta}(t) = \zeta_{i,j}(t)\}. \tag{3.3}$$

The restricted behavior $\mathfrak{B}_{0,j}(\zeta_{i,j}, [0, t_c])$ represents the set of all feasible futures at time $t = t_c$ for an RV tracing the SUT's trajectory. The expected severity for the restricted behavior indicates the criticality of the SUT's behavior at time t_c, under the jth parameter value.

We define the severity function, $\text{sev} : \mathfrak{B} \to \mathbb{R}^{\geq 0}$ such that the function is zero only for non-colliding trajectories, as only scenarios/trajectories with collisions of the SUT are considered to have non-zero severity. Furthermore, we consider $\mathfrak{B}_{0,j}(\zeta_{i,j}, [0, t_c])$ as our sample space and specify a cumulative distribution function $F(\zeta)$, to compute the expected value of severity function $\mathbb{E}[\text{sev}]$ and define:

$$R_p(t_c, \zeta_{i,j}) = \mathbb{E}[\text{sev}] = \int_{\mathfrak{B}_{0,j}(\zeta_{i,j},[0,t_c])} \text{sev}(\zeta) \mathrm{d}F(\zeta). \tag{3.4}$$

Evaluating $R_p(t_c, \zeta_{i,j})$ for each time instant t_c in a scenario $\zeta_{i,j} \in \mathfrak{B}_{i,j}$ yields the CM as a function of time $R_p(t, \zeta_{i,j})$. The detailed computation of $R_p(t, \zeta_{i,j})$ is discussed in Section 4.2 using an example.

3.3 Aggregation of CM for a scenario

Similar to [6], we define an aggregation of $R_p(t_c, \zeta_{i,j})$ over the complete scenario $\zeta_{i,j}$. Different signal norms for $R_p(t_c, \zeta_{i,j})$ can be chosen for this, depending on the application. In our application, if we are interested in the point of highest criticality in each scenario, we pick the infinity norm, i.e.,

$$\mathfrak{C}(\zeta_{i,j}) = \|R_p(t_c, \zeta_{i,j})\|_\infty = \max_{t_c \in \mathbb{T}_c} R_p(t_c, \zeta_{i,j}), \tag{3.5}$$

or the 1 norm if we are interested in including what happened over the entire scenario,

$$\mathfrak{C}(\zeta_{i,j}) = \|R_p(t_c, \zeta_{i,j})\|_1. \tag{3.6}$$

Thus, we define the time-dependent CM $R_p(t_c, \zeta)$ and its aggregation $\mathfrak{C}(\zeta)$ over the complete scenario.

4. Applying the criticality measure

This section discusses the procedure for computation of the CM $R_p(t_c, \zeta)$ and its aggregation $\mathfrak{C}(\zeta)$ for an example ADS in different scenarios.

4.1 Defining the System of Interest

Consider a system with two vehicles, e and a driving on the X-Y plane similar to the one in [6]. The subscript $p \in \{e, a\}$ denotes the corresponding vehicle, and the subscript $q \in \{x, y\}$ denotes the X or Y components. The Variables Of Interest (VOI) are the positions $s_{p,q}$, and velocities $v_{p,q}$ of both vehicles in the XY plane. Thus, the VOIs $\zeta = \{s_{p,q}, v_{p,q}\} \in \mathbb{W} = \mathbb{R}^8$ and the time vector for scenarios is $\mathbb{T} = [t_0, t_{\text{end}}] \subset \mathbb{R}^+ \cup \{0\}$.

A scenario is represented as $\zeta(t) = \{s_{p,q}(t), v_{p,q}(t)\} \in \mathfrak{B}_0 = \mathbb{T} \to \mathbb{W}$. Now we proceed with defining the set of scenarios, i.e., the behaviors. The behavior $\mathfrak{B}_0$ includes all admissible trajectories ζ, such that physical laws and limitations governing the motion of the RV are respected. For clarity of examples, the RV is constrained to 1-dimensional (1D) motion. The behavioral relations for $\mathfrak{B}_0$ are:

$$\text{Differentiability:} \quad s_{p,q}(t) \in C^2(\mathbb{T}, \mathbb{R}^4), \tag{4.1}$$

$$\text{By definition:} \quad \dot{s}_{p,q}(t) = v_{p,q}(t), \tag{4.2}$$

$$\text{Acceleration limits:} \quad a_{\min} \leq \dot{v}_{e,x}(t) \leq a_{\max}, \tag{4.3}$$

$$\text{No reverse driving:} \quad v_{e,x}(t) \geq 0, \tag{4.4}$$

$$\text{1D motion along } X\text{:} \quad v_{e,y}(t) = 0. \tag{4.5}$$

The set $\mathfrak{A}_j$ contains trajectories that satisfy the actor behavior parametrization and the initial conditions of the vehicles. The behavioral relations for $\mathfrak{A}_j$ are:

$$\text{Initial conditions:} \quad \zeta(0) = \{s_{p,q,0}, v_{p,q,0}\}, \tag{4.6}$$

$$\text{Actor behavior:} \quad \dot{v}_a(t) = 0 \tag{4.7}$$

In (4.6) and (4.7), $s_{p,q,0}$ and $v_{p,q,0}$ are constants making up the jth parameter set. The actor behavior and its parametrization are a choice of the SBT optimization setup. We consider non-interactive actors in the examples for ease of exposition, although the behavioral approach applies to both interactive and non-interactive actors. For interactive actors, the parameters may specify a model for the actor's interaction behavior.

The set $\mathfrak{S}_i$ contains all trajectories ζ which are permissible given the ith version of the ADS system. For the SUT, an Intelligent Driver Model (IDM) [10] longitudinal controller is implemented

as an ADS on a 2nd order kinematic point mass model representing the RV. The IDM is chosen due to its relative simplicity and ease of generating illustrative scenarios showcasing the CM. The constants, $\{c_{i,k} : k = \{1, ..., 5\}\}$ describes the ith version of the ADS with the behavioral relations:

$$\dot{v}_{e,x}(t) = a_1 + a_2 \tag{4.8}$$

$$\text{where,} \quad a_1 = 10\left(1 - \left(\frac{v_{e,x}(t)}{v_{e,x,0}}\right)^4\right), \tag{4.9}$$

$$a_2 = -c_{i,1}\left(\frac{c_{i,4} + c_{i,5}v_{e,x}(t)}{s_{e,x}(t) - \hat{s}_{a,x}(t)}\right)^2, \quad \text{and,} \tag{4.10}$$

$$\hat{s}_{a,x}(t) = \text{sense}\left(s_{p,q}(t)\right) = \begin{cases} s_{a,x}(t) \text{ ; Actor CG in FOV} \\ 1/\epsilon, \epsilon \to 0 \text{ ; Otherwise.} \end{cases} \tag{4.11}$$

The ADS has a perfect sensor with a limited Field of View (FOV) as shown in purple in Fig. 4.2 to detect the position of the other actor's Center of gravity (CG). Thus, we can write compactly,

$$\mathfrak{B}_0 = \{\zeta : \zeta \text{ satisfies (4.1) to (4.5)}\}, \tag{4.12}$$

$$\mathfrak{A}_j = \{\zeta : \zeta \text{ satisfies (4.6) to (4.7)}\}, \tag{4.13}$$

$$\mathfrak{S}_i = \{\zeta : \zeta \text{ satisfies (4.8) to (4.11)}\}. \tag{4.14}$$

4.2 Computing the measure

According to the previous section, given a scenario with the ith ADS version and jth parameter set $\zeta_{i,j} \in \mathfrak{B}_{i,j}$ and the RV's behavior, $\mathfrak{B}_{0,j}$, we now specify the steps of computing the CM $R_p(t_c, \zeta_{i,j})$ at a time step t_c and its aggregate $\mathfrak{C}(\zeta)$.

4.2.1 Defining the analysis time interval

We compute the criticality of a scenario at finite time instants $t_c \in \{t_1, t_2, t_3, ..., t_n\} = \mathbb{T}_c \subset \mathbb{T}$ and for each time instant $t_c \in \mathbb{T}_c$, consider an analysis interval as:

$$\mathbb{T}_{\text{pred}} = [t_c, t_c + t_h], \tag{4.15}$$

where t_h is the time horizon of analysis.

4.2.2 Defining the restricted behavior

The restricted behavior, defined in Section 3, is a subset of the behavior containing all trajectories that share values with the trajectory $\zeta_{i,j}$ in a time-frame $t \in [t_1, t_2] \subset \mathbb{T}$. For computing $\mathfrak{C}$ at $t = t_c$, we consider the restricted behavior, $\mathfrak{B}_{0,j}(\zeta_{i,j}, [0, t_c])$. For ease of readability, we define, $\mathfrak{B}_{\text{res}} = \mathfrak{B}_{0,j}(\zeta_{i,j}, [0, t_c])$.

4.2.3 Analysing the restricted set for severity

Considering all the trajectories in the restricted behavior, $\mathfrak{B}_{\text{res}}$, we aim to compute the expected value of severity. Assume $O_e(t)$ and $O_a(t)$ are the occupancy sets of the ego and actor respectively, which define the points (x, y) that lie within their respective bounding boxes at time t. Picking the absolute relative velocity at collision $|\Delta v_{\text{coll}}|$ to indicate severity, the severity function sev : $\mathfrak{B} \to \mathbb{R}^{\geq 0}$ is defined as,

$$\text{sev}(\zeta) = \begin{cases} |\Delta v_{\text{coll}}| & O_a(t) \cap O_e(t) \neq \emptyset, \text{Ego collides} \\ 0 & \text{Otherwise.} \end{cases} \tag{4.16}$$

The choice of the severity function sev, among the many others in the literature, is arbitrary, as the focus is on showcasing how the method can be applied.

Furthermore, each scenario $\zeta \in \mathfrak{B}_{\text{res}}$ has a probability of occurrence. We consider sev as a random variable on the sample space of trajectories and define its expected value:

$$\mathbb{E}[\text{sev}] = \int_{\mathfrak{B}_{\text{res}}} \text{sev}(\zeta) \mathrm{d}F(\zeta), \tag{4.17}$$

where $F(\zeta)$ is a cumulative distribution function defined over the space of scenarios. Monte-Carlo (MC) sampling can be used to approximate the value of $\mathbb{E}[\text{sev}]$ by generating/sampling trajectories in $\mathfrak{B}_{\text{res}}$ as given in,

$$R_p(t_c, \zeta_{i,j}) = \mathbb{E}[\text{sev}] \approx \frac{1}{N_s} \sum_{k=1}^{N_s} \text{sev}(\zeta_k), \quad \zeta_k \leftarrow F. \tag{4.18}$$

Since the set $\mathfrak{B}_{\text{res}}$ is infinite dimensional, we define a subset of this space $\mathfrak{B}_{\text{res,pc}} \subset \mathfrak{B}_{\text{res}}$ containing all trajectories generated by piece-wise constant acceleration inputs which is an approximation of $\mathfrak{B}_{\text{res}}$ to generate samples from and evaluate $\mathbb{E}[\text{sev}]$. The samples ζ_k must be drawn based on the probability distribution of trajectories.

The probability distribution of trajectories can be estimated from data or based on heuristic models. Specifying a valid distribution lies outside the scope of this work, and the focus is on showcasing how probability distributions can be incorporated into the framework for computing the CM. For this paper, we consider each scenario in $\mathfrak{B}_{\text{res,pc}}$ to be parametrized by the piece-wise constant input sequence, $U = (u_{i=1}, u_2 ... u_{n\text{step}}) \in \mathbb{R}^{n\text{step}}$, where the input u_i at each time step is sampled from the cumulative distribution function of the Singer model [1] shown in Figure 4.1. The forward acceleration inputs $a_f(t)$ to the RV are correlated in time [1] and described by, $\dot{a}_f = -(u_i(t) - a_f(t))/\tau$, where τ is the maneuver time constant. We pick $\tau = 0.1s$, and higher values of τ indicate slower changing input, i.e., slower maneuvers. The probability of the input sequence U is the probability of the generated scenario $\zeta \in \mathfrak{B}_{\text{res,pc}}$. Note that the choice of the Singer model as a probability distribution of inputs is not scenario dependent.

To sample trajectories in $\mathfrak{B}_{\text{res,pc}}$, we discretize the timespan $\mathbb{T}_{\text{pred}}$ (4.15) into n_{step} equal steps and generate N_s samples of input sequences for the Monte Carlo approximation of $\mathbb{E}[\text{sev}]$. The initial conditions for the RV are the position and velocity of the SUT at $t = t_c$ in the scenario.

4.2.4 Aggregated CM for a scenario

Picking the appropriate signal norm for $R_p(t_c, \zeta)$ provides the criticality for the complete scenario $\mathfrak{C}(\zeta)$. This choice depends on the application, and sample choices of the ∞ norm and the 1 norm are discussed in the examples.

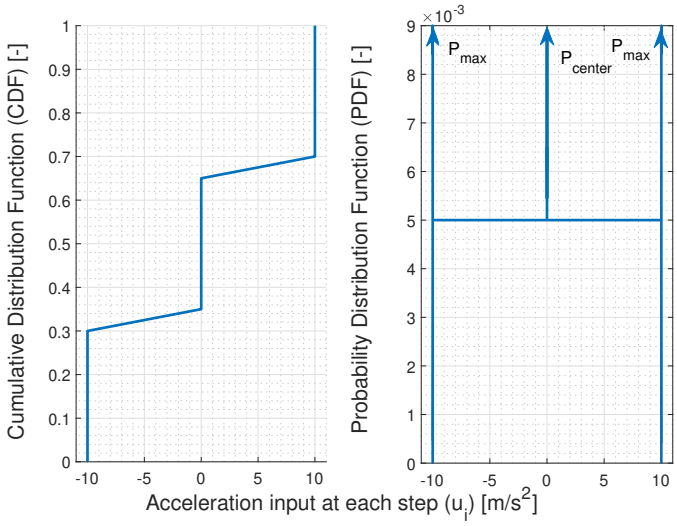

Figure 4.1: Cumulative distribution function and probability density function of acceleration inputs (Singer model [1]).

4.3 Results and discussion

This section contains an example of an intersection scenario set to which the proposed CM has been applied. The example scenarios A, B, and C described below are identical to those presented in Kurian et al. [6], while Scenario G is additional. The resulting values for $R_p(t_c, \zeta_{i,j})$ and its aggregation $\mathfrak{C}(\zeta)$ are discussed.

4.3.1 Intersection scenario: Crossing actor

Consider an intersection with two vehicles as shown in Fig. 4.2. Dotted lines depict the paths followed by vehicles a and e. The swept area of the ego vehicle as it travels from left to right is shown in yellow, and purple represents the field of view of the sensor mounted on the ego vehicle. A perfect sensor detects the position and velocity of the actor's CG within the field of view, and within a lane width $l_w = 6.5m$. The actor and ego vehicle are initialized at $(s_{a,x,0}, s_{a,y,0}) = (40, -15)m$

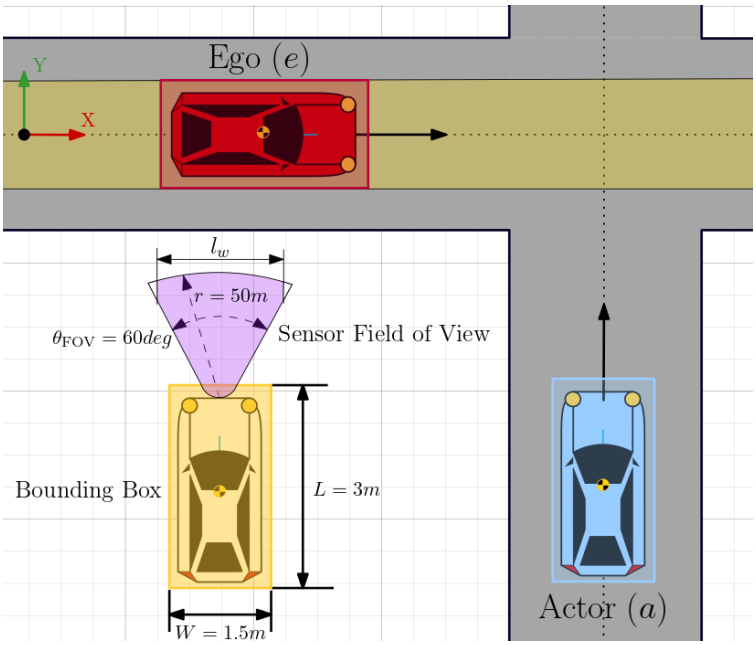

Figure 4.2: Intersection scenario: Bounding boxes and layout.

and $(s_{e,x,0}, s_{e,y,0}) = (0, 0)m$ respectively. The initial velocity of the actor and ego along their path is $v_{a,x,0} = 4.5m/s$ and $v_{e,x,0}$ respectively. Scenario parameters, $c_{i,4} = 1.5m$ and $c_{i,5}$, listed in Table 4.1, represent the standstill distance and headway time maintained by the IDM controller, respectively. Different values for $c_{i,5}$ and $v_{e,x,0}$ are chosen to change the behavior of the ego vehicle and influence scenario criticality.

Scenario ID	A	B	C	G
Headway time $c_{i,5}$	0.5 s	4.0 s	0.0 s	0.5 s
Ego initial speed $v_{e,x,0}$	10.3 m/s	10.3 m/s	10.3 m/s	13 m/s

Table 4.1: Parameters for scenarios A, B, C, and G.

Consider four scenarios, A, B, C, and G, for the system above. The scenario parameters are given in Table 4.1, and the position-time plots for the CG of the ego vehicle in A, B, C, and G are shown in Fig. 4.3. The shaded blue region depicts the occupancy of actor a in the ego vehicle's path during the crossing. The extent of the ego vehicle's bounding box along the x direction is shown with dashed lines. The crossing actor a is within the FOV and detected by the ego vehicle at $t = 2.6s$. Beyond this, each IDM controller responds with different braking decelerations to avoid colliding with a. In A and B, the decelerations are sufficient to avoid a collision, as opposed to C and G, where a collision occurs.

Fig. 4.4 shows a portion of the position-time plot for scenario B. To show the computation of CM at the time instant of analysis $t_c = 1s$, the plot is overlaid with multiple future trajectories of the RV's CG generated within its physical limits. For visualization, Fig. 4.4 shows the trajectories generated with $N_s = 15$. For all examples, the time horizon for analysis is chosen as $t_h = 3s$, and the trajectories are generated with $N_s = 1000$ and $n_{\text{step}} = 15$. The acceleration range for the RV is between $[a_{f,\min}, a_{f,\max}] = [-10, 10]\, m/s^2$. Each trajectory is evaluated for the severity of collision with the occupancy of the actor and is colored accordingly. The occupancy of the actor is independent of whether the ego vehicle has detected the actor at $t = t_c$ as the analysis is not from the ego point of view. The resulting $R_p(t)$ for scenarios A, B, and C, shown in Fig. 4.5 is discussed in Section 4.3.2.

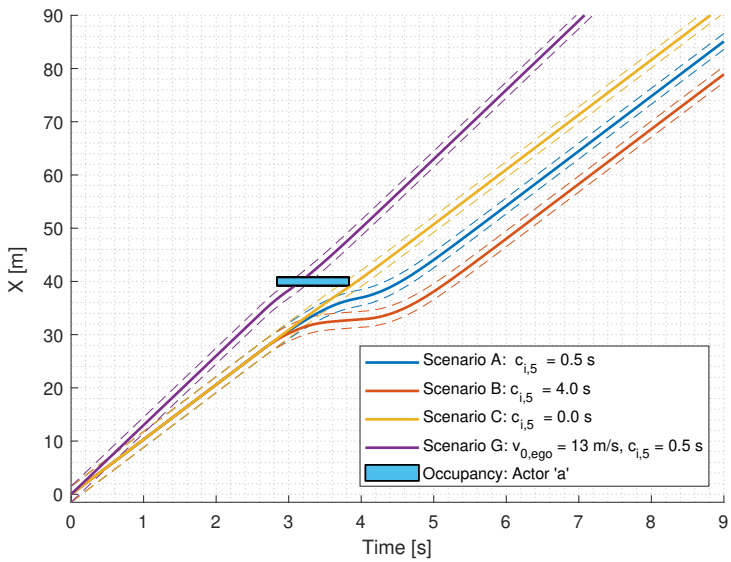

Figure 4.3: Intersection scenario: $s_{e,x}$ vs. t plot.

4.3.2 Discussion

For scenarios A, B, C, and G, the plot of R_p vs. t_c is shown in Fig. 4.5. The small variations in $R_p(t)$ arise from the Monte Carlo approximation in Eq. 4.18. For scenarios C and G involving a collision, the computation is only valid until the collision shown in fig 4.5, which is sufficient for our application. As vehicle e approaches the actor a, $R_p(t_c, \zeta)$ for A, B, and C increases similarly until $t_c = 2.6s$, i.e., before e detects a, as the position-time plots are identical.

$R_p(t_c, \zeta)$ for scenario G is higher than the other scenarios due to the higher approach velocity

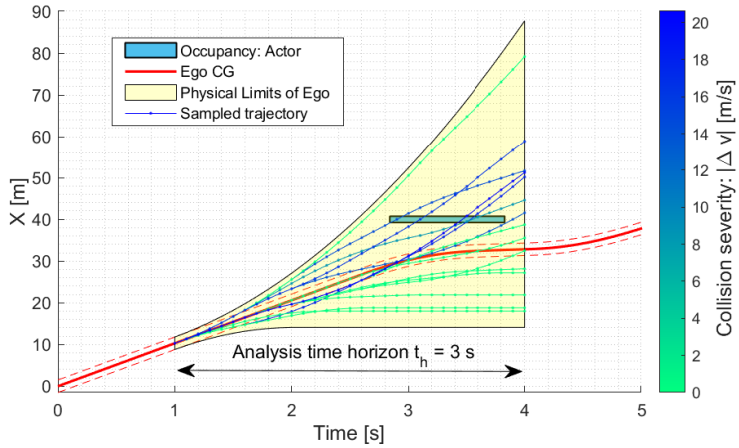

Figure 4.4: Computing $R_p(t_c, \zeta)$ at $t_c = 1s$.

leading to higher collision risk. Thus, inclusion of the severity aspect helps distinguish the two collision scenarios, C and G. This is key to defining scenario hazardousness and evaluating risk from unknown-unsafe scenarios.

Considering the aggregation of $R_p(t_c, \zeta)$ over the complete scenario as the peak value, i.e., $\mathfrak{C}(\zeta) = ||R_p(t_c, \zeta)||_\infty$, we find scenarios A and B are evaluated as identical based on their point of highest criticality, followed by scenario C and scenario G respectively. On the other hand, if we consider a different aggregation, i.e., $\mathfrak{C}(\zeta) = ||R_p(t_c, \zeta)||_1$, we find A and B are evaluated differently, with A shown as being more critical than B. A and B are near-miss scenarios and $\mathfrak{C}(\zeta) = ||R_p(t_c, \zeta)||_1$ distinguishes between them, similar to the measure presented in [6].

For scenarios A and B, the increase in the value of $R_p(t_c, \zeta)$ with the peak occurring at the point when the IDM starts braking is intuitive due to the IDM braking upon detecting the vehicle. An IDM with a lower value for headway time $c_{i,5}$, with other parameters kept constant, is prone to higher collision risk. From Fig. 4.3, after $t_c = 2.6s$ we can see that the collision risk for e follows the order C>A>B, and the values of $R_p(t_c, \zeta)$ show this. Thus, $R_p(t_c, \zeta)$ measures the collision risk and matches with intuition for scenario criticality.

The aggregated value of $R_p(t_c, \zeta)$ includes the probability and severity aspect to distinguish between two colliding scenarios based on collision risk, aligning with intuitive results about a scenario's criticality. It also retains the properties of the measure of 'proximity to collisions' from Kurian et al. [6], such as sensitivity to near-misses. Lastly, it incorporates the controllability and severity aspect of ISO 26262 [11] by considering the physical limits of the RV in its computation. Revisiting the requirements from Section 2.1, we see that $R_p(t_c, \zeta)$ and its aggregation $\mathfrak{C}(\zeta)$ satisfies R4 in addition to R1, R2, R3-a, and R3-b.

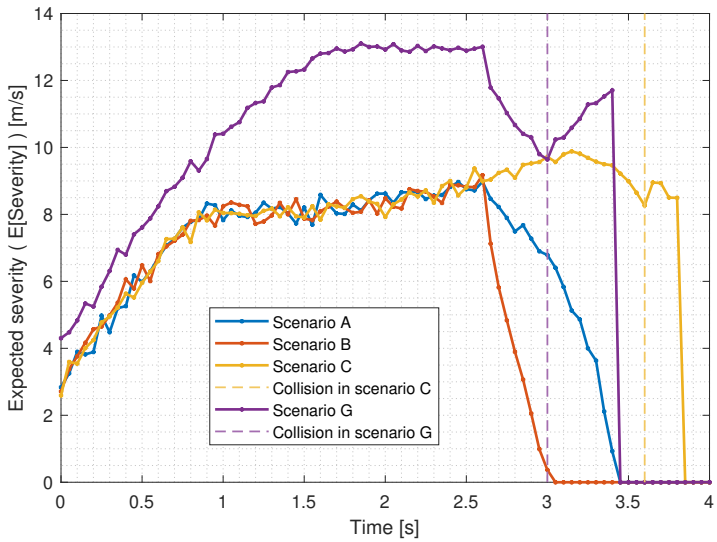

Figure 4.5: Criticality of intersection scenario: $R_p(t_c, \zeta)$ vs. t_c.

5. Conclusions

The main goal of this deliverable is to quantify the risk of unknown-unsafe scenarios. To find unknown-unsafe scenarios in accordance with ISO 21448 using optimization, there is a need for an objective function, i.e., a scenario CM that meets certain requirements. Existing measures in the literature fail to satisfy these requirements and this work proposes an improved scenario risk/criticality measure meeting these requirements. The proposed CM includes the probability and severity aspect and is interpretable as 'proximity to highly probable, severe collisions'. The behavioral modeling approach is used to describe the proposed measure.

The proposed measure is based on the collision risk, i.e., expected severity, associated with an uncontrolled Reference Vehicle (RV) tracing the trajectory of the SUT. The CM assumes a specified behavior for the actors and is computed retrospectively for a scenario, including information about the future actor motions. The CM is showcased by applying it to example scenarios at an intersection. It satisfies the requirements of a cost function and matches the intuitive expectation of scenario criticality. Thus, the CM, when integrated as the objective function for the optimization problem of SBT, can help generate and evaluate unknown high-risk scenarios as part of the work in WP5.

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